



FACULTY OF SCIENCE – UNIVERSITY OF PERADENIYA
DEPARTMENT OF STATISTICS AND COMPUTER SCIENCE
END SEMESTER EXAMINATION (SEMESTER I) – 2019/2020
ST103 – STATISTICS APPLICATIONS I (1 CREDIT)

Time allowed: **Two hours**

Answer **ALL** questions

INSTRUCTIONS: You should prepare a text document (MS – WORD or Writer) including all the necessary outputs, answers and relevant interpretation. Save the file with the name **XXXXXX.doc** or **XXXXXX.odt** where **XXXXXX** represents your registration number (Eg: If the registration number is “S/18/001” then the file name should be “**S18001.doc**” or “**S18001.odt**”)

Also save the R-studio project with the relevant R-scripts with the name “**XXXXXX.Rproj**” in your home directory. **XXXXXX** represents your registration number. Inside this project, create a separate R-script file for each question including all the R commands and save your R-script files with the file names below.

Question	1	2	3	4
R-script file	Que1.R	Que2.R	Que3.R	Que4.R

You are advised to save all your works regularly to avoid losses of data due to power failures.

1. Consider the text file **Covid19** in your home directory to answer the following questions. This dataset was obtained from the Covid vaccination program conducted by the World Health Organization and consists of nine variables as follows.

Variable	Description
Country	Country for which the vaccination information is provided
Code	ISO code for the country
Date	Date for the data entry
Total_vaccinations	Absolute number of total immunizations in the country
People_vaccinated	Number of people that received immunization according to the immunization scheme (1 or 2 vaccines)
People_fully_vaccinated	Number of people that received the entire set of immunization according to the immunization scheme (typically 2 vaccines)
Daily_vaccinations	Number of people vaccinated per day
Vaccines	Type of vaccines used in the country
Source_name	Source of the information (national authority, international organization, local organization etc.)

- i. Import the dataset as **Covid_vaccines**
- ii. Using a suitable R command, change the column names **Code** and **Vaccine** to **ISO_Code** and **Vaccine_type**.

- iii. Obtain the summary statistics for **People_fully_vaccinated**, **Daily_vaccinations** and interpret the results separately.
- iv. Create a new vector called **People_partially_vaccinated** by subtracting the variable **People_fully_vaccinated** from **People_vaccinated**, and create a new data frame called **Covid_vaccines_new** by combining the **People_partially_vaccinated** vector to the **Covid_vaccines** data frame.
- v. Select the details of Covid vaccination programs conducted in the United Kingdom (United Kingdom consists of four countries: England (GBR), Scotland, Wales and Northern Ireland). Store the results in a new data frame called **UK_vaccination_Details**.
- vi. Create a contingency table for the two categorical variables **ISO_Code** and **Vaccine_type**.
- vii. Draw a horizontal boxplot for the variable **Daily_vaccinations** according to the **Country**.
- viii. Find the frequency of **Source_name** and create a pie chart for the same variable using rainbow colours. Give a suitable title and display the source name and the frequency as the labels of the pie chart.

2. a)

- i. Create two sequences **seq_1** and **seq_2** by including odd and even numbers (from 0 to 20) as follows.

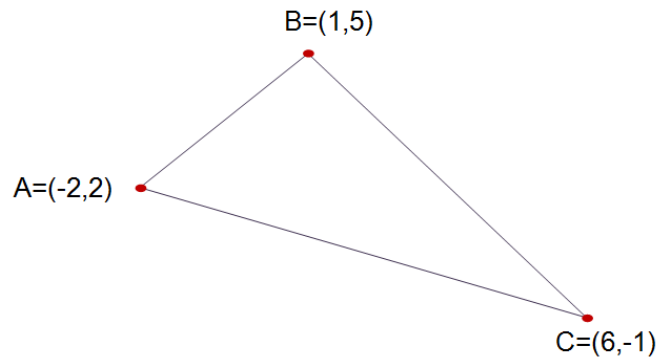
$$seq_1 = \{19, 17, \dots, 3, 1\} \text{ and } seq_2 = \{20, 18, \dots, 2, 0\}$$
- ii. Create 2×2 matrices **A** and **B** with the first 4 elements of **seq_1** and **seq_2**, respectively (enter the elements row-wise).
- iii. Do the matrices **A** and **B** have an inverse? Justify your answer.
- iv. Find **A'**, **B'**, **2A + B**, **A - 3B** and **AB** (both element wise multiplication and matrix multiplication)
- v. Find a matrix **C** such that **CA = 2B**. Round off the elements of **C** to a whole number.
- vi. Create a 2×2 diagonal matrix **I** (identity matrix) with the value 1 in its diagonal.
- vii. Prove the fact that **A⁻¹A = I = AA⁻¹**.
Hint: You can use *all.equal(X, Y)* function to check if X and Y are equal.

- b) Consider a triangle with vertices at (x_1, y_1) , (x_2, y_2) , and (x_3, y_3) . The area of this triangle can be found using the determinant of the matrix **D** as given in the following formula.

$$Area = \pm \frac{1}{2} |D|, \text{ where } D = \begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix}.$$

Here, the plus or minus sign of the answer should be taken such that final answer (area of the triangle) is non-negative.

Consider the ABC triangle given below and answer the questions i to viii.



- i. Create a 3×2 empty matrix **mat_A**.
 - ii. Create two vectors **x_vec** and **y_vec** by including the x and y coordinates of the vertices of the ABC triangle, respectively.
 - iii. Fill the first and second column of **mat_A** using the values of **x_vec** and **y_vec**, respectively.
 - iv. Add another column to **mat_A** where all the elements are equal to 1.
 - v. Find the dimension of **mat_A**.
 - vi. Change the column names of **mat_A** to **x**, **y** and **z**. Change the row names as **A**, **B** and **C**.
 - vii. Find the determinant of **mat_A** and divide it by 2.
 - viii. What is the area of the ABC triangle?
Hint: You can use $abs(x)$ function to get the absolute value of x
3. a) Create the below given sequence of vectors using the functions **seq** and **rep** when necessary.
- i. $\{3, 6, 9, \dots, 24\}$
 - ii. $\{28, 24, 20, \dots, 4\}$
 - iii. $\{2, 4, 6, \dots, 14, 16, 14, 12, \dots, 2\}$
 - iv. $\{3, 3, \dots, 3, 5, 5, \dots, 5, 7, 7, \dots, 7\}$, where there are 5 occurrences of 3, 10 occurrences of 5 and 15 occurrences of 7.
 - v. $\{11, 13, 15, 11, 13, 15, \dots, 11, 13, 15\}$, where there are 10 occurrences of the set $\{11, 13, 15\}$
- b) Create two data frames named **Employee_Details** and **IQ_Results** and fill them with

the following observations.

Employee_Details:

Emp. Number	Name	Gender
21001	Jim	M
21002	Mary	F
21003	Arthur	M
21004	Natali	F
21005	Emma	F

IQ_Results:

Emp. Number	Test 1	Test 2	Test 3
21001	56	61	78
21002	86	82	84
21003	64	67	60
21004	90	93	82
21005	45	34	57

- i. Merge **Employee_Details** and **IQ_Results** datasets and name the merged dataset as **Merge_data**
 - ii. Update Jim's Test 1 results to 75 and Arthur's Test 2 results to 81.
 - iii. Extract a subset named **Female_Emp** from **Merge_data** such that it contains all the records of female employees (Gender=F)
 - iv. Plot a line graph for the three tests and give a suitable title, x-axis name and y-axis name.
 - v. Place a legend (indicating the corresponding test) at the top-right corner with a frame and using the same colours.
4. a) A particular project has three phases. There are two types of costs associated with each phase of the project. They are the fixed cost which is independent of its duration, and the variable cost which depends on the duration. The following table summarizes the costs associated with each phase of the project.

PHASE	1	2	3
Fixed Cost (\$)	318 000	212 000	220 000
Variable Cost (\$ per day)	15 000	14 000	16 000

Let x , y and z be the required number days to cover each phase. That implies,

$$Cost_{phase1} = 318\,000 + 15\,000x$$

$$Cost_{phase2} = 212\,000 + 14\,000y$$

$$Cost_{phase3} = 220\,000 + 16\,000z$$

$$Total\ Cost = Cost_{phase1} + Cost_{phase2} + Cost_{phase3}$$

The designer of the project must propose a price for the project. He would like to set a price that ensures a profit margin of at least 10%.

$$Proposed\ Price = 1.1 \times Total\ Cost$$

- i. Write a function to express the total cost of the project and the price the designer should propose. You should use x, y and z as initial arguments. The output should contain ***Total cost*** and the ***Proposed price***.
- ii. Calculate the total cost and proposed price when $x = 5, y = 6, z = 2$
- b) Write an R program to print the output of the following recursive function:

$$f(x, y) = \begin{cases} 1 & ; y = 0 \\ xf(x, y - 1) & ; y > 0, \\ xf(x, -y) & ; y < 0 \end{cases}$$

where x, y are integers. Run the function when $x = 5$ and $y = 4$.

[End of Paper]